Module 6 -- Robotics Projects

Lesson 46: Obstacle-Avoiding Robot

★ Star-5 Expert | ⏰ 50--60 minutes | 👤 Age 6--12 years | ⚡ 350 XP

Your Mission Today

Robot Navigator, your robot can follow a line — but what happens when something is in the way? Today you are going to give your robot the power of **echolocation** — just like a bat! You will mount an ultrasonic distance sensor on the front and use a servo to scan left and right. When the robot sees an obstacle, it will stop, look around, and choose the best path — all by itself. This is **autonomous obstacle avoidance**, and it is one of the most important skills in all of robotics!

Learning Objectives

By the end of this lesson, you will be able to:

- Explain how ultrasonic distance sensors measure distance
- Mount the HC-SR04 sensor and scanning servo on the robot
- Write an obstacle-avoidance algorithm
- Build an obstacle course and test your robot
- Use your Magic Measurement Wand to measure ultrasonic sensor VCC and trigger pin voltages

What You Need

Item	Qty
Your assembled robot from Lessons 43-44	1
HC-SR04 ultrasonic distance sensor	1
SG90 servo motor	1
Servo sensor mounting bracket (or cardboard + hot glue)	1
Jumper wires	8
Cardboard boxes or books (for obstacles)	5+
Multimeter (your Magic Measurement Wand!)	1
USB cable (for programming)	1

How to Teach This Lesson

Step 1: Hook — The Bat Robot (5 min)

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> "Close your eyes and try to walk across the room without bumping into anything."

(Let the kid try — gently guide them away from real danger.)

💬

> "Hard, right? Now imagine doing that using SOUND. Bats do this every night! They send out a high-pitched squeak, and when the sound bounces back from a wall or a tree, the bat knows how far away it is. This is called ECHOLOCATION."

Hold up the HC-SR04 sensor:

> "This sensor does the EXACT same thing! It sends out an ultrasonic pulse — a sound so high that humans cannot hear it — and listens for the echo. The longer it takes for the echo to come back, the farther away the object is. Your robot is about to become a bat!"

```
How Ultrasonic Sensing Works:

SENSOR ))))))))  ---------> WALL
                             |
SENSOR ((((((((  <--------- ECHO

Time between send and receive = distance!
Distance = (time x speed of sound) / 2
```

Award: **+10 XP** **for understanding echolocation!**

Step 2: Mounting the Sensor and Servo (10 min)

The scanning system:

The HC-SR04 sensor is mounted on top of the servo motor. The servo can rotate the sensor left and right to scan for obstacles in different directions.

```
Scanning System (Side View):

     [HC-SR04]         <-- ultrasonic sensor (eyes)
        |
     [Servo]           <-- rotates the sensor
        |
 =================     <-- robot chassis
 O               O     <-- wheels
```

Assembly:

1. Attach the servo horn (arm) to the servo motor
2. Mount the HC-SR04 on the servo horn using a bracket, or make one from cardboard:

```
DIY Sensor Mount:

+-------[HC-SR04]-------+
|                        |
+---+              +---+
    |              |
    +---[HORN]---+
         |
      [SERVO]
```

1. Mount the servo on the FRONT of the robot chassis (centered)
2. Make sure the sensor faces forward when the servo is at 90 degrees

Wiring:

```
HC-SR04:                    Servo:
VCC ---> Arduino 5V         Red   ---> Arduino 5V
GND ---> Arduino GND        Brown ---> Arduino GND
Trig --> Arduino Pin 11     Signal --> Arduino Pin 6
Echo --> Arduino Pin 12
```

Award: **+20 XP** **for mounting the scanning system!**

Step 3: Testing the Distance Sensor (5 min)

First, let us make sure the sensor works:

```
// Lesson 46: Distance Sensor Test
int trigPin = 11;
int echoPin = 12;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600);
  Serial.println("Distance Sensor Test");
}

void loop() {
  // Send ultrasonic pulse
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Read echo
  long duration = pulseIn(echoPin, HIGH);
  int distance = duration * 0.034 / 2;

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  delay(200);
}
```

Test it:

- Hold your hand in front of the sensor and move it closer/farther
- Check that the Serial Monitor readings match the actual distance (roughly)
- Try measuring a wall, a book, your hand

> 💬
>
> "If the readings jump around wildly, make sure nothing is blocking the sensor's path and that the two 'eyes' (transmitter and receiver) are clean."

Award: **+20 XP** **for getting accurate distance readings!**

Step 4: Testing the Servo Scan (5 min)

```
// Servo scan test
#include <Servo.h>
Servo scanServo;

void setup() {
  scanServo.attach(6);
  Serial.begin(9600);
}

void loop() {
  Serial.println("Looking LEFT");
  scanServo.write(0);     // Look left
  delay(1000);

  Serial.println("Looking CENTER");
  scanServo.write(90);    // Look center
  delay(1000);

  Serial.println("Looking RIGHT");
  scanServo.write(180);   // Look right
  delay(1000);

  Serial.println("Looking CENTER");
  scanServo.write(90);    // Back to center
  delay(1000);
}
```

"Watch your sensor head turn left, center, right, center. Like a robot looking around the room! This is exactly what the robot will do when it finds an obstacle."

Award: **+10 XP** **for testing the servo scan!**

Step 5: The Obstacle-Avoidance Algorithm (12 min)

Here is the FULL autonomous obstacle-avoidance code:

```
// Lesson 46: Obstacle-Avoiding Robot
#include <Servo.h>

// Motor pins
int leftENA = 9;
int leftIN1 = 2;
int leftIN2 = 3;
int rightENB = 10;
int rightIN3 = 4;
int rightIN4 = 5;

// Ultrasonic sensor pins
int trigPin = 11;
int echoPin = 12;

// Servo
Servo scanServo;
int servoPin = 6;

// Settings
int driveSpeed = 170;
int turnSpeed = 180;
int dangerDistance = 25;  // cm -- stop if closer than this

void setup() {
  // Motor pins
  pinMode(leftENA, OUTPUT);
  pinMode(leftIN1, OUTPUT);
  pinMode(leftIN2, OUTPUT);
  pinMode(rightENB, OUTPUT);
  pinMode(rightIN3, OUTPUT);
  pinMode(rightIN4, OUTPUT);

  // Sensor pins
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

  // Servo
  scanServo.attach(servoPin);
  scanServo.write(90);  // Start looking forward

  Serial.begin(9600);
  Serial.println("Obstacle Avoider Ready!");
  delay(2000);
}
```

```
void loop() {
  int distance = getDistance();

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  if (distance > dangerDistance) {
    // Path is clear -- drive forward!
    forward(driveSpeed);
  } else {
    // OBSTACLE DETECTED! Stop and scan.
    Serial.println("OBSTACLE! Scanning...");
    stopRobot();
    delay(300);

    // Look LEFT
    scanServo.write(10);
    delay(500);
    int leftDist = getDistance();
    Serial.print("Left distance: ");
    Serial.println(leftDist);

    // Look RIGHT
    scanServo.write(170);
    delay(500);
    int rightDist = getDistance();
    Serial.print("Right distance: ");
    Serial.println(rightDist);

    // Look back to CENTER
    scanServo.write(90);
    delay(300);

    // Choose the direction with MORE space
    if (leftDist > rightDist && leftDist > dangerDistance) {
      Serial.println("Turning LEFT -- more space");
      turnLeft(turnSpeed);
      delay(500);
    } else if (rightDist > dangerDistance) {
      Serial.println("Turning RIGHT -- more space");
      turnRight(turnSpeed);
      delay(500);
    } else {
      // Both sides blocked! Back up and try again
```

```
      delay(800);
      turnRight(turnSpeed);  // Turn around
      delay(700);
    }
    stopRobot();
    delay(200);
  }

  delay(50);
}

// === DISTANCE FUNCTION ===
int getDistance() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH, 30000);  // 30ms timeout
  if (duration == 0) return 999;  // No echo = very far
  return duration * 0.034 / 2;
}

// === MOVEMENT FUNCTIONS ===
void forward(int speed) {
  digitalWrite(leftIN1, HIGH);
  digitalWrite(leftIN2, LOW);
  digitalWrite(rightIN3, HIGH);
  digitalWrite(rightIN4, LOW);
  analogWrite(leftENA, speed);
  analogWrite(rightENB, speed);
}

void backward(int speed) {
  digitalWrite(leftIN1, LOW);
  digitalWrite(leftIN2, HIGH);
  digitalWrite(rightIN3, LOW);
  digitalWrite(rightIN4, HIGH);
  analogWrite(leftENA, speed);
  analogWrite(rightENB, speed);
}

void turnLeft(int speed) {
  digitalWrite(leftIN1, LOW);
```

```
  digitalWrite(rightIN4, LOW);
  analogWrite(leftENA, speed);
  analogWrite(rightENB, speed);
}

void turnRight(int speed) {
  digitalWrite(leftIN1, HIGH);
  digitalWrite(leftIN2, LOW);
  digitalWrite(rightIN3, LOW);
  digitalWrite(rightIN4, HIGH);
  analogWrite(leftENA, speed);
  analogWrite(rightENB, speed);
}

void stopRobot() {
  digitalWrite(leftIN1, LOW);
  digitalWrite(leftIN2, LOW);
  digitalWrite(rightIN3, LOW);
  digitalWrite(rightIN4, LOW);
}
```

Walk through the logic:

```
Obstacle Avoidance Flowchart:

START
  |
  v
Measure distance ahead
  |
  v
Distance > 25cm? ---YES---> Drive Forward
  |                              |
  NO                             |
  |                              |
  v                              |
STOP!                            |
  |                              |
  v                              |
Scan LEFT -- record distance     |
Scan RIGHT -- record distance    |
  |                              |
  v                              |
Left > Right?                    |
  YES --> Turn LEFT              |
  NO ---> Turn RIGHT             |
  |                              |
  v                              |
Both blocked?                    |
  YES --> Back up + turn around  |
  |                              |
  +------------------------------+
```

Award: **+40 XP** **for uploading the obstacle-avoidance code!**

Step 6: Build the Obstacle Course (5 min)

Set up an obstacle course using cardboard boxes, books, or other objects:

```
Example Obstacle Course:

START
  |
  v
+---+        +---+
|BOX|        |BOX|
+---+   +    +---+
        |
     +--+---+
     | BOOK  |
     +-------+
                   +---+
          +---+    |BOX|
          |BOX|    +---+
          +---+
                      ^
                      |
                    FINISH
```

Tips:

- Leave enough space between obstacles for the robot to fit through
- Start with wide gaps and make them tighter as the robot improves
- Use objects that the ultrasonic sensor can detect (solid surfaces, not fabric)

Award: **+20 XP** **for building an obstacle course!**

Step 7: Tuning and Testing (5 min)

Let the robot run through the course! Adjust these values:

Parameter	Effect	Try This
dangerDistance	How close before stopping	15—30 cm
driveSpeed	How fast it drives	140—200
Turn delay	How much it turns	300—700 ms
Backup delay	How far it reverses	500—1000 ms

> 💬
> "Your robot is making DECISIONS. It sees an obstacle, thinks about which way has more space, and chooses the best path. This is the beginning of artificial intelligence!"

Award: **+20 XP** **for a successful obstacle-course run!**

Step 8: Wand Check — Ultrasonic Sensor Voltages (8 min)

> 💬
> "Time for your Magic Measurement Wand to check the sensor signals!"

Set the Wand to **DC Volts**.

Measurement 1: Sensor Power

```
| Measurement              | Expected | Your Reading |
|--------------------------|---------|-------------|
| HC-SR04 VCC pin          | 5.0V     |             |
| HC-SR04 GND pin          | 0V       |             |
| Servo VCC (red wire)     | 5.0V     |             |
```

Measurement 2: Trigger Pin Voltage

The Trig pin sends a quick pulse. With the robot running, measure the Trig pin. You might see a fluctuating reading since the pulse is very fast:

```
| Measurement                | Expected    | Your Reading |
|----------------------------|------------|-------------|
| Trig pin (idle)            | ~0V         |             |
| Trig pin (while measuring) | Flickers 0-5V |           |
```

> "The trigger pulse is only 10 microseconds long — that is 10 MILLIONTHS of a second! Your Wand might not catch it perfectly because it is SO fast, but you should see it flicker. This is one of the fastest signals you have measured!"

Measurement 3: Servo Signal Wire

Measure the voltage on the servo signal wire while it moves:

```
| Servo Position | Signal Voltage (approx) |
|----------------|------------------------|
| 0 degrees      |                        |
| 90 degrees     |                        |
| 180 degrees    |                        |
```

> "The servo signal is also a PWM pulse — your Wand will show an average voltage that changes with the position. Pretty cool!"

Award: **+50 XP** **for all Wand measurements!**

Quick Quiz — Earn Bonus XP!

Question 1: How does the HC-SR04 ultrasonic sensor measure distance?

- A) It uses a camera to see objects
- B) It sends a sound pulse and measures how long the echo takes to return
- C) It measures temperature changes

(Correct: B — **+20 XP** **!)**

- A) To make the sensor spin faster
- B) To scan left and right and find which direction has more open space
- C) To make the robot look cool

(Correct: B — +20 XP !)

Question 3: The robot detects an obstacle 15cm ahead. Left scan reads 50cm. Right scan reads 10cm. Which way should it turn?

- A) Left — because 50cm is more space
- B) Right — because 10cm is closer
- C) Straight ahead

(Correct: A — +20 XP !)

Lesson 46 Complete!

```
==============================================

   OBSTACLE NAVIGATOR BADGE UNLOCKED!

   Skills unlocked:
   [check] Mounted ultrasonic sensor + servo
   [check] Wrote distance measurement function
   [check] Programmed obstacle-avoidance logic
   [check] Built and tested obstacle course
   [check] Measured sensor voltages with Wand

==============================================
```

XP Breakdown:

Activity	XP
Hook (echolocation)	10
Mounting scanning system	20
Distance sensor test	20
Servo scan test	10
Obstacle-avoidance code	40
Build obstacle course	20
Successful course run	20
Wand Check (all measurements)	50
Quiz (3 questions)	60
TOTAL POSSIBLE	**250**

Coming Up Next...

In **Lesson 47**, your robot gets a PERSONALITY! You will add LEDs for eyes and a buzzer for a voice. Your robot will flash its lights when turning, beep when it spots an obstacle, and play a startup melody when it powers on. Get ready to make your robot come alive!

Troubleshooting

Problem	Fix
Distance readings are 0 or 999	Check Trig and Echo pin connections
Sensor always reads very short distances	Something blocking sensor face — check mounting
Servo jitters or does not move	Check 5V power — servo might need more current
Robot crashes into obstacles	Increase dangerDistance or reduce driveSpeed
Robot gets stuck in corners	Increase backup time in the "TRAPPED" section of code
Wand reads 3.5V on HC-SR04 VCC	Power supply weak — check batteries